

Arthur Gontier

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Doctor in Computer Science

Profile: I am a post-doc researcher with experiences in the fields of constraint programming, explanations, automated proofs, proof analysis, and broader knowledge of optimisation and cryptanalysis. I am interested in solver upgrades especially in the area of automated proofs, explanation and learning.

Formation and Experiences.

- **Post-Doc: Proof producing constraint solvers** 2024-2026
University of Glasgow *School of Computing Science*
 - **Collaborators:** Ciaran MCCREESH, Matthew MCILREE, and the VeriPB group: veripb.org
- **PhD: Mathematical programming and cryptanalysis** 2020-2023
Rennes University 1 *Computer science*
 - **Position:** I did my PhD in the EMSEC and Capsule teams at the IRISA laboratory. My director was Stephanie Delaune and my supervisors were Patrick Derbez and Charles Prud'homme.
 - **Title:** Cryptanalysis of symmetric cipher using generic solvers
 - **Speciality:** Computer Science
 - **Jury:**
 - * Christina BOURA, Reviewer , Senior lecturer, Université de Versailles Saint-Quentin-en-Yvelines
 - * Stéphanie DELAUNE, examiner, Research Director at CNRS, Rennes
 - * Patrick DERBEZ, examiner, Senior lecturer, Rennes University 1
 - * Thomas PEYRIN, Reviewer, Professor, Nanyang Technological University of Singapore
 - * Charles PRUD'HOMME, examiner, Senior lecturer, IMT Atlantique, Nantes
 - * Christine SOLNON, defence director, Professor, INSA Lyon
 - **Research:** In this PhD I studied the creation and security of symmetric ciphers with optimisation tools such as Linear, Boolean and Constraint solvers.
 - **Teaching:** During these three years I gave lectures to various levels. From the basic programming courses at the first year bachelors level to linear programming and graph theory at the master level.
- **Master: Optimisation and operational research** 2018-2020
Nantes University and Université Libre de Bruxelles
 - **Internships:** The first year I did a research internship on multi-objective optimisation in the SLP team of the LS2N laboratory at the University of Nantes. The second year did a six month internship in the TASK team (LS2N) at the IMT Atlantique on a research project for constraint programming solver.
 - **ERASMUS:** I made half a year at the Université Libre de Bruxelles to get a double graduation.

- **Bachelor: Mathematics and Computer Science** 2015-2018
Nantes University *Valedictorian*
 - **Internship:** I did an three month internship on the subject of compiling and static analysis at École Normale Supérieure of Lyon in the team CASH of the Laboratoire de l'Informatique du Parallélisme.

Teachings

- **Graph theory** 2020
Renne University 1 , Bachelor 3
- **Integer Linear Programming** 2020
Rennes University 1 , Master 1
- **Functional programming** 2021
INSA Rennes, Master 1
- **Computer Science** 2021-2023
Rennes University 1, Bachelor 1
- **Constraint programming (examiner)** 2025
Glasgow University 1, Master 1

Publications

- **Faster Certified Symmetry Breaking** AAAI 2026
- **Using Orders With Auxiliary Variables**
 - **Authors:** Markus Anders, Bart Bogaerts, Benjamin Bogø, Arthur Gontier, Wietze Koops, Ciaran McCreesh, Magnus O. Myreen, Jakob Nordström, Andy Oertel, Adrian Rebola-Pardo, Yong Kiam Tan.
 - **Type:** 26 pages in an international peer-reviewed conference, rank: A1
 - **Abstract:** Symmetry breaking is a crucial technique in modern combinatorial solving, but it is difficult to be sure it is implemented correctly. The most successful approach to deal with bugs is to make solvers certifying, so that they output not just a solution, but also a mathematical proof of correctness in a standard format, which can then be checked by a formally verified checker. This requires justifying symmetry reasoning within the proof, but developing efficient methods for this has remained a long-standing open challenge. A fully general approach was recently proposed by Bogaerts et al. (2023), but it relies on encoding lexicographic orders with big integers, which quickly becomes infeasible for large symmetries. In this work, we develop a method for instead encoding orders with auxiliary variables. We show that this leads to orders-of-magnitude speed-ups in both theory and practice by running experiments on proof logging and checking for SAT symmetry breaking using the state-of-the-art SATSUMA symmetry breaker and the VERIPB proof checking toolchain.
 - **Contributions:** For this work, I was involved in the design of the proof system auxiliary variables features. I developed the proof trimming prototype and I was also involved in the benchmark and complexity analysis of the results.
- **A CP-based automatic tool for instantiating truncated differential characteristics** INDOCRYPT 2023

- **Authors:** François Delobel, Patrick Derbez, Arthur Gontier, Loïc Rouquette, Christine Solnon.
- **Type:** 40 pages in an international peer-reviewed conference, rank: B
- **Abstract:** An important criteria to assert the security of a cryptographic primitive is its resistance against differential cryptanalysis. For word-oriented primitives, a common technique to determine the number of rounds required to ensure the immunity against differential distinguishers is to consider truncated differential characteristics and to count the number of active S-boxes. Doing so allows one to provide an upper bound on the probability of the best differential characteristic with a reduced computational cost. However, in order to design very efficient primitives, it might be needed to evaluate the probability more accurately. This is usually done in a second step, during which one tries to instantiate truncated differential characteristics with actual values and computes its corresponding probability. This step is usually done either with ad-hoc algorithms or with CP, SAT or MILP models that are solved by generic solvers. In this paper, we present a generic tool for automatically generating these models to handle all word-oriented ciphers. Furthermore the running times to solve these models are very competitive with all the previous dedicated approaches.
- **Contributions:** For this project, I worked with members of the Lyon team and developed the core of the model generator with Loïc Rouquette. Subsequently, I proposed optimizations such as parallel competitive computing and graph simplification. I also wrote most of the paper and presented the work at the conference.

- **New algorithm for exhausting optimal permutations
for generalized feistel networks**

INDOCRYPT 2022

- **Authors:** Stéphanie Delaune, Patrick Derbez, Arthur Gontier, and Charles Prud'homme.
- **Type:** 22 pages in an international peer-reviewed conference, rank: B
- **Abstract:** The Feistel construction is one of the most studied ways of building block ciphers. Several generalizations were proposed in the literature, leading to the Generalized Feistel Network (GFN) construction, in which the round function operates on each pair of blocks in parallel until all branches are permuted. At FSE'10, Suzaki and Minematsu studied the diffusion of such construction, raising the question of how many rounds are required so that each block of the ciphertext depends on all blocks of the plaintext. Exhausting all possible permutations up to 16 blocks, they observed that there were always optimal permutations mapping even-number input blocks to odd-number output blocks and vice versa. Recently, both Cauchois et al. and Derbez et al. proposed new algorithms to build optimal even-odd permutations for up to 36 blocks. In this paper, we present a new algorithm based on iterative path building to search for optimal Feistel permutation. This algorithm is much faster in exhausting optimal non-even-odd permutations than all the previous approaches. Our first result is a computational proof that no non-even-odd permutation reaches a better diffusion round than optimal even-odd permutations up to 32 blocks. Furthermore, it is well known that permutations with an optimal diffusion round do not always lead to optimal permutations against differential cryptanalysis. We investigate several new criteria to build permutations leading to more secure GFN.
- **Contributions:** For this publication, I carried out all the work: formalization, implementation, as well as writing and presentation at the conference. My co-authors provided me with feedback at each of these stages.

- **A simpler model for recovering superpoly on trivium**
Selected Areas of Cryptography

SAC 2021

- **Authors:** Stéphanie Delaune, Patrick Derbez, Arthur Gontier, and Charles Prud'homme.

- **Type:** 20 pages in an international peer-reviewed conference, rank: B
- **Abstract:** The cube attack is a powerful cryptanalysis technique against symmetric primitives, especially for stream ciphers. One of the key step in a cube attack is recovering the superpoly. The division property has been introduced to cube attacks with the aim first to identify variables/monomials that are not involved in the superpoly. Recently, some improved versions of this technique allowing the recovery of the exact superpoly have been developed and applied on various stream ciphers. In this paper, we propose a new model to recover the exact superpoly of a stream cipher given a cube. We model the polynomials involved in the stream cipher as a directed graph. It happens that this structure handles some of the monomial cancellations more easily than those based on division property, and this leads to better timing results. We propose two implementations of our model, one in MILP and one in CP, which are up to 10 times faster than the original division property-based model from Hao et al. , and consistently 30 to 60 times faster than the monomial prediction-based model from Hu et al.
- **Contributions:** For this publication, I formalized the problem, developed two of the three implementations, wrote the technical section of the paper, and presented it at the conference. My co-authors provided feedback at each of these stages.

- **Conflict analysis in CP solving:**

CPTAI 2020

- **Explanation generation from constraint decomposition**

- **Authors:** Arthur Gontier, Charles Prud'homme and Charlotte Truchet.
- **Type:** 7 pages in an international peer-reviewed workshop.
- **Abstract:** Constraint programming solvers can use a clause learning technique inspired from CDCL. However, this method relies on an explanation method for each constraint used in the problem. Manually generating these constraint explanations can be lengthy and tedious, given that there are several hundred constraints in the constraint catalog. In this work, we propose a rewriting algorithm to generate these explanations and demonstrate how to use it for about ten constraints. The algorithm is based on constraint decompositions, meaning that a constraint can be rewritten into several smaller, equivalent constraints. We have proposed a minimalist decomposition language and provided rewriting rules for each of its constraints. From the decomposition of a global constraint in this small language, our algorithm can deduce the explanations of global constraint. This publication shows that it is possible to generate these explanations but also raises the question of their quality. Indeed, if the decomposition of the global constraint does not have good filtering, then the explanation generated from this decomposition will not be very effective either.
- **Contributions:** For this publication, I carried out all the work: formalization, implementation, as well as writing and presentation at the conference. My co-authors provided me with feedback at each of these stages.

Invited Talks and small publications

- **A variety of trimming techniques for pseudo-Boolean proof logs** WOOPS 2025
 - **Type:** Presentation of Pseudo Boolean proof trimming research advances at the veriPB workshop.
 - **Abstract:** Informal presentation of the latest PB proof trimming researches.
- **Pseudo-Boolean proof trimming** WOOPS 2024
 - **Type:** Presentation of Pseudo Boolean proof trimming early techniques at the veriPB workshop.

- **Abstract:** Informal presentation of early PB proof trimming.
- **Génération d’explications de contraintes par leurs décompositions** ROADEF 2023
 - **Type:** 2 pages + presentation at the French conference of operational research.
 - **Abstract:** Informal presentation of the work later published in the CPTAI 2020 paper.
- **Diffusion totale dans le schéma de Feistel généralisé** ROADEF 2022
 - **Type:** 2 pages + presentation at the French conference of operational research.
 - **Abstract:** Informal presentation of the work later published in the INDOCRYPT 2022 paper.

Skills

Programming

Julia, Ocaml, C++, Rust, Java, Python, Latex, html

Languages

French

English

Optimisation methods

CP, MILP, SAT, SMT and parallel Ad-hoc Algorithms

Others

Secretary of a symphonic Orchestra

(Concert organisation)

Global Game Jam 2016

(Video game creation in 48h)